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Inhaltsverzeichnis

Abbildungsverzeichnis

Tabellenverzeichnis

Abstract

Eine Kurzzusammenfassung der Vorgehensweise und der wesentlichen Ergebnisse.

Allgemeine Merkmale

- Objektivität: Es soll sich jeder persönlichen Wertung enthalten.
- Kürze: Es soll so kurz wie möglich sein.
- Verständlichkeit: Es weist eine klare, nachvollziehbare Sprache und Struktur auf.
- Vollständigkeit: Alle wesentlichen Sachverhalte sollen enthalten sein.
- Genauigkeit: Es soll genau die Inhalte und die Meinung der Originalarbeit wiedergeben.

1 Einleitung

2 Literature Review

2.1 Maintenance Optimization in Infrastructure Systems

Maintenance optimization in infrastructure systems addresses the challenge of planning, scheduling, and executing maintenance activities to ensure system availability, reliability, and cost-efficiency over the lifecycle of technical assets. Unlike manufacturing or logistics contexts, infrastructure maintenance often involves geographically distributed assets, limited maintenance resources, and significant consequences of system failures (de Jonge & Scarf, 2020; Dekker, 1996).

This chapter provides the conceptual foundation for maintenance optimization. Section 2.1.1 defines maintenance optimization and its objectives. Section 2.1.2 reviews maintenance strategies: preventive, reactive, and condition-based. Section 2.1.3 discusses uncertainty, availability, and cost as central decision variables. Section 2.1.4 provides the transition to mobile maintenance and service interventions (de Jonge & Scarf, 2020; Alaswad & Xiang, 2017).

2.1.1 Concept and Objectives of Maintenance Optimization

Maintenance optimization refers to the systematic determination of when and what to maintain in order to achieve specific objectives such as minimizing costs, maximizing availability, or balancing both (de Jonge & Scarf, 2020). The fundamental trade-off lies between the cost of maintenance activities and the cost of system failures or downtime (Dekker, 1996).

Early work by Dekker (1996) identified six barriers preventing the application of maintenance optimization models in practice: models are difficult to understand, many papers are written only for mathematical purposes, maintenance activities consist of many different aspects, optimization is not always necessary, models often focus on the wrong type of maintenance, and there is a gap between theory and operational reality. Despite these challenges, maintenance optimization remains strategically important because continuous developments of technical systems and increasing reliance on equipment have resulted in growing importance of effective maintenance activities (de Jonge & Scarf, 2020).

In the context of charging infrastructure, maintenance optimization must address not only individual station failures but also fleet-level dispatching, resource allocation, and timing decisions under uncertainty (de Jonge & Scarf, 2020). The objective is typically to minimize total expected costs (maintenance plus failure costs) while maintaining a required level of system availability (Alaswad & Xiang, 2017; Dekker, 1996).

2.1.2 Maintenance Strategies: Preventive, Reactive, Condition-Based

Maintenance strategies can be broadly classified into three categories: reactive (corrective) maintenance, preventive (planned) maintenance, and condition-based (predictive) maintenance (Sarja & Halilov, 2023; de Jonge & Scarf, 2020).

Reactive maintenance (also called corrective or run-to-failure maintenance) is performed only after a failure occurs (Sarja & Halilov, 2023). This strategy is suitable for non-critical assets where downtime has minimal impact on operations. However, for critical infrastructure such as charging stations, reactive maintenance can lead to significant customer dissatisfaction, safety risks, and higher repair costs due to unanticipated failures (de Jonge & Scarf, 2020).

Preventive maintenance is performed according to a fixed schedule or usage-based interval, regardless of the actual condition of the asset (Sarja & Halilov, 2023; Alaswad & Xiang, 2017). Preventive maintenance aims to prevent failures before they occur by replacing or servicing components at predetermined times. This approach is well-established and widely applied, for example in vehicle maintenance or periodic inspections of technical systems (Dekker, 1996). However, preventive maintenance can lead to over-maintenance (replacing components that are still functional) or under-maintenance (failures occurring before the scheduled interval) (de Jonge & Scarf, 2020).

Condition-based maintenance (CBM) performs maintenance based on the actual condition of the asset, which is monitored through sensors, inspections, or performance data (Alaswad & Xiang, 2017; Sarja & Halilov, 2023). CBM enables maintenance to be performed exactly when needed, reducing unnecessary interventions and focusing resources on assets that actually require attention. In the literature, CBM is often extended to predictive maintenance, which uses data analytics, machine learning, or digital twins to predict when maintenance will be needed (Sarja & Halilov, 2023).

The review by Alaswad and Xiang (2017) on condition-based maintenance optimization models for stochastically deteriorating systems shows that CBM is particularly suitable for systems where condition monitoring is feasible and failure consequences are significant. For charging infrastructure, this includes monitoring of charging power, error rates, temperature, or communication status (de Jonge & Scarf, 2020).

In practice, many organizations use a hybrid strategy, combining preventive maintenance for critical components with reactive maintenance for non-critical ones and condition-based maintenance for assets where monitoring is available (Sarja & Halilov, 2023; de Jonge & Scarf, 2020). For this thesis, the hybrid perspective is relevant because maintenance of charging infrastructure often involves both planned inspections and unplanned failure responses.

2.1.3

Maintenance decisions are inherently made under uncertainty. Key uncertain parameters include failure times, service times, travel times, availability of spare parts, and arrival of new maintenance requests (de Jonge & Scarf, 2020; Dekker, 1996). This uncertainty affects three central decision variables:

Availability measures the proportion of time that a system is operational and able to fulfill its function (Sarja & Halilov, 2023). For charging infrastructure, availability is a key performance indicator because failed stations directly reduce customer access and revenue. Predictive maintenance can increase availability by 5–15 % according to McKinsey estimates, while reducing maintenance costs by 18–25 % (Sarja & Halilov, 2023).

Cost includes maintenance costs (labor, materials, travel), failure costs (downtime, customer dissatisfaction, penalties), and inventory costs (spare parts, tools) (de Jonge & Scarf, 2020; Dekker, 1996). The optimization objective is typically to minimize total expected cost over a planning horizon, subject to availability constraints (Alaswad & Xiang, 2017).
Uncertainty affects both the timing and the impact of maintenance decisions. For example, a failure may occur earlier or later than expected, service may take longer than planned, or travel to a site may be delayed by traffic (de Jonge & Scarf, 2020). Decision-making under uncertainty follows a sequential process in which uncertainty realization (observation of the true values of previously uncertain parameters) and subsequent reactive decision-making alternate as time proceeds (de Jonge & Scarf, 2020).
In maintenance optimization, uncertainty is often modeled using stochastic deterioration processes, probability distributions for failure times, or scenario-based approaches (Alaswad & Xiang, 2017; de Jonge & Scarf, 2020). The choice of uncertainty model depends on the amount of available data, the complexity of the system, and the computational requirements of the optimization approach.

2.1.4 Transition to Mobile Maintenance and Service Interventions

The maintenance problems discussed in Sections 2.1.1–2.1.3 typically focus on when and what to maintain at individual assets. However, in many practical settings, maintenance is performed by mobile technicians who must travel between geographically distributed sites, which introduces routing and scheduling decisions (de Jonge & Scarf, 2020).
For charging infrastructure maintenance, this means that the optimization problem is not only about determining Maintenance intervals or failure thresholds but also about dispatching technicians, planning routes, and allocating limited resources (vehicles, spare parts, time) across multiple stations (de Jonge & Scarf, 2020; Alaswad & Xiang, 2017).
This transition from single-asset maintenance optimization to mobile maintenance service planning introduces additional complexity:
• Spatial dependency: Maintenance decisions for one station affect the availability of resources for other stations (de Jonge & Scarf, 2020). • Temporal dependency: The order in which stations are visited affects total travel time and service availability (Alaswad & Xiang, 2017). • Uncertainty coupling: Failures, travel times, and service times are often stochastic and interdependent (de Jonge & Scarf, 2020). The literature on maintenance optimization has traditionally focused on single-asset or multi-asset models without explicit routing (de Jonge & Scarf, 2020; Dekker, 1996). More recent work begins to integrate maintenance planning with vehicle routing, especially in the context of dynamic and stochastic vehicle routing problems (SDVRP) (de Jonge & Scarf, 2020). This integration is precisely the focus of the following chapters 2.2 and 2.3, which cover stochastic routing and approximate dynamic programming for dynamic service planning.
For this thesis, the key insight from Chapter 2.1 is that maintenance optimization provides the decision objectives (cost, availability, uncertainty handling), while Chapters 2.2 and 2.3 provide the methodological framework (SDVRP, ADP) for implementing these objectives in a mobile, dynamic, and stochastic service environment (de Jonge & Scarf, 2020; Alaswad

& Xiang, 2017).

3 Problem Definition

This chapter introduces the practical planning problem addressed in the thesis: the maintenance and disturbance handling of a network of electric vehicle charging stations. The purpose of the chapter is to describe the operational environment, the sources of uncertainty, the relevant assumptions, and the cost structure in a way that motivates the later mathematical abstraction. The formal Markov decision process and the approximate dynamic programming framework are therefore not introduced here, but are prepared conceptually by clarifying the real-world planning context.

The problem considered in this thesis involves the daily routing and scheduling of mobile maintenance teams responsible for servicing a geographically distributed network of charging stations. The planning problem is not static: in addition to regular preventive tasks, new disturbance events may occur during the day and require operational responses on short notice. As a result, the system combines elements of preventive maintenance, corrective maintenance, and dynamic routing.

The decision process is organized on a daily basis. At the beginning of each day, an initial service plan is constructed for both teams. During the operating day, this plan may be revised in response to newly occurring faults. The detailed structure of the daily planning cycle, the types of disturbances encountered, and the cost model underlying the optimization problem are described in the following subsections.

From a modeling perspective, the chapter establishes the practical structure of the system before any formal abstraction is introduced. In particular, it clarifies that the planning problem is not merely a vehicle routing problem with fixed customer requests, but a dynamic maintenance problem in which service requirements evolve over time, operational conditions change throughout the day, and routing decisions must be adapted repeatedly.

3.1 Operational Setting

The operational scenario considered in this thesis is based on the charging infrastructure of the city of Würzburg, consisting of 397 stations distributed across the urban service area. These charging stations form the service network that must be maintained by two mobile service teams. Each team starts its route at a central depot and is required to return to the depot by the end of the working day. The service region is represented as a graph whose node set consists of the depot and all charging stations. Travel between stations is therefore a core element of the problem, since every maintenance action requires physical movement through the network.

The planning horizon is structured as a daily operational process. The working day runs from 8:00 to 17:00. At the beginning of the day, a full initial plan is generated. During the day, replanning decisions are triggered event-driven on an hourly grid whenever new disturbance events arrive. This means that routing decisions are not made once and then executed unchanged. Instead, they are revised as new information becomes available. Such a planning logic is essential in practice because maintenance operations in charging infrastructure are rarely fully predictable at the start of the day.

A further operational characteristic is the presence of a deterministic lunch break between 12:00 and 13:00. In the route-based setup, this break is not treated as an independent decision variable. Instead, it is inserted according to a fixed operational rule and shifts all subsequent arrival and departure times accordingly. This reflects the idea that some daily routines are known in advance and need not be optimized explicitly, but still influence route feasibility and time consumption.

The spatial structure of the system also plays an important role. Because the charging stations are distributed across the service region, route efficiency depends strongly on travel distances, travel times, and geographic clustering. For this reason, the operational logic uses zones as a practical pre-structuring device. The stations are grouped into zones which support the initial selection of promising service areas for the two teams. These zones reduce routing complexity and guide the daily assignment of work areas, though the formal zone selection mechanism is introduced in detail in a later chapter.

Overall, the operational setting illustrates that the problem is a daily service planning task under operational constraints rather than a long-term static assignment problem. The interaction between limited team capacity, geographically dispersed stations, fixed working hours, and dynamically emerging disturbances creates a planning environment in which route design and real-time adaptation are equally important.

3.2 Environment and Disturbances

A defining feature of the planning problem is the presence of uncertainty. The maintenance teams operate in an environment in which service requirements are not fully known at the start of the day. New faults may occur while teams are already executing their planned routes, and the operational system therefore evolves over time. For this reason, a purely static routing approach is insufficient. Instead, the planning logic must account for the fact that the set of required service actions can change during execution.

In this setting, disturbances are treated as exogenous operational events. They occur stochastically and may emerge at different charging stations at different times of the day. Once a disturbance occurs, it immediately creates an additional service requirement that competes for limited maintenance capacity. This means that service planning must continuously balance already scheduled preventive work with newly arriving corrective tasks. The problem is therefore not only about finding efficient routes, but also about deciding how to reallocate attention when the system state changes unexpectedly.

The route-based model distinguishes between two types of disturbances. Type-1 faults are simple failures that can be resolved through standard repair actions with a fixed service time. Type-2 faults are more complex and require disassembly, a depot roundtrip, and reassembly, resulting in a significantly longer and station-dependent service time. This distinction is operationally important because the insertion of a short Type-1 job and the insertion of a long Type-2 repair do not have the same impact on the remainder of the day.

Uncertainty is not limited to the occurrence of disturbances. Travel conditions also vary over time and are subject to uncertainty. Travel times depend on the time of day due to varying traffic conditions, and their actual realization is stochastic. This means that geographically similar routes may still differ substantially in execution quality depending on temporal traffic patterns and local congestion effects.
Another relevant feature of the environment is that unresolved disturbance jobs do not disappear at the end of the day. If a newly occurring fault cannot be inserted feasibly into the remaining route of either team, it is transferred as a carryover job to the following day. These carryover disturbances are then prioritized in the next daily planning cycle. This mechanism introduces an intertemporal dependency into the operational system: today's capacity shortage affects tomorrow's workload. In practical terms, the planning problem therefore includes not only immediate routing consequences but also backlog management over multiple days.

3.3 Assumptions and Cost Model

To make the planning problem analytically tractable, the operational system is represented through a number of simplifying assumptions. First, not every real-world process is modeled in full physical detail. Some side processes associated with complex repairs are incorporated through aggregate service-time components instead of explicit intermediate route legs. In the current formulation, a Type-2 fault includes additional time components for disassembly, depot-related handling, and reassembly, but the team remains conceptually assigned to the affected station rather than physically traveling to and from the depot as a separate route segment. This simplification reduces model complexity while preserving the main operational time impact of such events.
Second, service requirements are represented as operational jobs rather than as fully persistent technical condition states. Open regular maintenance tasks, new faults, and carryover disturbances are treated as actionable job sets. In addition, each charging station is associated with a maintenance-related status through the indicator of whether periodic maintenance has been completed and through the variable days since maintenance, which later drives failure risk. However, the binary distinction between "healthy" and "faulty" infrastructure is not carried as a separate persistent state variable in the operational problem description. This abstraction supports a cleaner transition into the route-based MDP in Chapter 4.
The economic evaluation of the planning problem is based on three cost components. The first component consists of labor costs for the maintenance teams, set to 35 euros per hour. These costs capture the personnel resources committed to route execution. The second component consists of travel costs, set to 0.30 euros per kilometer, which reflect the logistical burden of moving between geographically distributed stations. The third component consists of downtime-related costs, set to 0.50 euros per kilowatt-hour, which quantify the service impact of unresolved disturbances. Together, these three components reflect that good operational performance depends not only on efficient routing, but also on timely fault resolution.

The cost structure creates a clear trade-off. A route with low travel distance is not necessarily desirable if it leaves important faults unserved for too long. Conversely, a strongly service-oriented plan that aggressively inserts every disturbance may lead to high labor and travel expenditure or may jeopardize the timely return to the depot. The relevant operational objective is therefore not the minimization of a single physical quantity such as distance, but the minimization of total expected cost across labor use, mobility effort, and service delay consequences.

This chapter has established the real-world planning setting, the uncertainty structure, and the economic logic underlying the maintenance problem. These elements provide the basis for the formal model in the next chapter. Chapter 4 translates the operational setting into a route-based Markov decision process and then connects that formulation to an approximate dynamic programming solution approach.

At the same time, this chapter deliberately stops short of a full state-based technical representation of each charging station. The operational environment is described here at a conceptual level: faults occur, they generate service requirements, and they influence routing decisions. The precise mathematical representation of station-specific maintenance status, days since maintenance, and stochastic failure generation is introduced only in Chapter 4. This separation is intentional, because Chapter 3 is concerned with explaining the real decision environment, not yet with formalizing it.

4 Theoretical Framework

This chapter develops the formal framework used to represent the maintenance routing problem as a route-based Markov decision process. Building on the operational setting introduced in Chapter 3, the objective is now to translate the planning problem into a mathematical structure that captures sequential decision-making under uncertainty. Particular emphasis is placed on the interaction between route construction, stochastic fault arrivals, time-dependent travel conditions, and limited daily service capacity. The chapter further explains why a state-action-value perspective is required and why approximate dynamic programming constitutes an appropriate solution methodology for the resulting problem.

4.1 Markov Decision Process

5 Solution Methodology

6 Experimental Setup

7 Results and Analysis

8 Discussion

9 Outlook and Future Work

A Anhang A

Hiermit versichere ich, die vorliegende Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie die Zitate deutlich kenntlich gemacht zu haben.
Ich erkläre weiterhin, dass die vorliegende Arbeit in gleicher oder ähnlicher Form noch nicht im Rahmen eines anderen Prüfungsverfahrens eingereicht wurde.

Würzburg, den 18. Juni 2026 VORNAME NACHNAME